6th International Innovation Competition
(ITC-EGYPT 2026)
Graduation Projects' Template

Faculty/school: Faculty of Computers & Artificial Intelligence

University / Studying administration: Modern University for Technology and Information (MTI)

Project Title:
[Smart Education Assist]

Supervisor:

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Executive Summary

Smart Education Assist is an intelligent educational platform designed to improve student engagement and enhance teaching effectiveness using artificial intelligence. The project addresses a critical problem in modern education, especially in Egypt: low student attention levels during lectures (especially in the virtual, online) and the lack of real-time feedback for teachers to evaluate student understanding.

Studies indicate that student attention significantly drops after the first 10–15 minutes of a lecture, leading to reduced knowledge retention and poor academic performance. Additionally, teachers often rely on subjective observation, which makes it difficult to accurately identify struggling or disengaged students.

To solve this problem, Smart Education Assist provides a **data-driven engagement analysis system** using computer vision and machine learning. The system analyzes visual cues such as face orientation and eye focus, combined with interaction data (e.g., quiz responses and participation), to generate a real-time **engagement score** for each student. A teacher dashboard presents these insights in a clear and actionable format, allowing instructors to adapt their teaching methods instantly.

A working **prototype** of the system is planned using a webcam-based setup, where student engagement is detected (e.g., eye direction detection, etc) through AI models built with Python (OpenCV and SVM). The frontend interface is developed using Flutter, while the backend is powered by .NET Core. The system also includes an automated post-lecture quiz module to evaluate student understanding and reinforce learning.

The project is practical and can be implemented using existing technologies with low-cost requirements, as it only depends on standard devices such as laptops with cameras and internet connectivity. This makes it suitable for deployment in schools, universities, and online learning platforms across Egypt.

Currently, the project is developed by a student team without external funding. However, it has strong potential for partnerships with educational institutions, EdTech companies, and government sectors.

Smart Education Assist contributes to the education industry by introducing **AI-driven decision-making in classrooms**, shifting from traditional passive teaching to interactive and adaptive learning environments. It enhances both teaching quality and student performance through measurable insights.

The system can be transformed into a scalable product offered as a **Software-as-a-Service (SaaS)** platform, allowing institutions to subscribe and integrate it into their existing systems. This creates strong potential for commercialization and startup development.

The project is based on scientific principles from:

- Artificial Intelligence (Machine Learning, Computer Vision)
- Human-Computer Interaction
- Educational Psychology (attention and active learning theories)

The main system components include:

- **Software:** AI models (Python), backend (.NET Core), frontend (Flutter), SQL databases
- **Hardware:** Camera-enabled devices
- **Technologies:** WebRTC, REST APIs, cloud infrastructure

The primary beneficiaries are:

- Teachers (better classroom insights)
- Students (improved learning experience)
- Educational institutions (higher performance and efficiency)

From a business perspective, the project has a clear model based on **subscription plans for schools and universities**, with additional premium analytics features. The initial development cost is minimal, mainly limited to development resources and basic infrastructure.

Future work will focus on improving AI accuracy, integrating natural language processing for analyzing student questions, and expanding the system to support hybrid and large-scale educational environments.

Team Structure and Responsibilities

Name	Contribution (%) and Description
Mohamed Ismail Fawzy Harras	25% – Team Leader and Backend Developer. Responsible for system architecture design, backend development using .NET Core, API development, database design, and overall project coordination and integration between different modules. Email: mohamedismailfh@gmail.com Phone Number: +20 11 40017805
Sherif Mohamed Mahmoud Elgendy	20% – Full-Stack Developer. Responsible for developing the backend APIs, and ensuring smooth user experience across the platform. Email: sherifelgendy2004@gmail.com Phone Number: +20 12 82952786
Youssef Harby Mohamed Sorur	20% – Mobile Application Developer (Flutter). Responsible for building the mobile version of the platform, ensuring cross-platform compatibility, and integrating mobile features with backend services. Email: youssef.sorur789@gmail.com Phone Number: +20 11 55682419
Abdelrahman Sami Abd Elmaqsoud	15% – UI/UX Designer and Frontend Support. Responsible for designing user interfaces, improving user experience, and assisting in frontend/mobile implementation to ensure usability and visual consistency. Email: as0144549@gmail.com Phone Number: +20 10 97457086
Youssef Ibrahim Soliman Mohamed	20% – AI/ML Engineer. Responsible for developing and training machine learning models, implementing computer vision algorithms (e.g., face and attention detection), and integrating AI components into the system. Email: youssefibrahim.ai@gmail.com Phone Number: +20 10 29377961

Partners and financiers.

At the current stage, **Smart Education Assist** is a student-led project developed independently without external financial support or official partnerships. The project is being designed and implemented by the team using available resources and tools.

However, the system has strong potential for future collaboration with:

- Educational institutions (schools and universities)
- E-learning platforms
- Governmental education sectors
- EdTech companies

These partnerships could support large-scale deployment, data-driven improvements, and integration with existing educational systems.

From a financial perspective, the project is currently self-funded with minimal development costs. In future stages, potential funding sources may include:

- Educational grants
- Startup incubators and accelerators
- Private investors interested in EdTech solutions

Abstract

Smart Education Assist is an intelligent educational platform designed to address the critical issue of low student engagement and lack of real-time feedback in modern classrooms. In both physical and online learning environments, teachers often struggle to accurately assess student attention and understanding during lectures, leading to reduced learning effectiveness and delayed intervention for struggling students.

To tackle this problem, Smart Education Assist proposes a data-driven approach that combines computer vision and machine learning to analyze student engagement in real time. The system utilizes non-intrusive visual cues such as face orientation and eye focus, along with interaction data like quiz responses and participation, to generate an engagement score for each student. These insights are presented through a dynamic teacher dashboard, enabling instructors to adapt their teaching strategies instantly.

The main contribution of this work is the development of an integrated system that bridges the gap between teaching and real-time student feedback, transforming traditional classrooms into adaptive learning environments. Additionally, the platform includes an automated post-lecture quiz module to reinforce learning and evaluate comprehension.

A functional prototype has been designed using technologies such as Python (OpenCV, SVM), Flutter, and .NET Core, demonstrating the feasibility of the system with minimal hardware requirements. The results indicate that AI-driven insights can significantly enhance teaching efficiency and student interaction.

In conclusion, Smart Education Assist presents a scalable and practical solution with strong potential for deployment in educational institutions and future commercialization as an EdTech product.

Abbreviations and Terms

Abbreviation	Definition
ML	Machine Learning – a subset of AI that enables systems to learn and improve from experience without being explicitly programmed.
CV	Computer Vision – a field of AI that enables machines to interpret and process visual data from images or videos.
UI/UX	User Interface / User Experience – the design and interaction experience of users with the system.
API	Application Programming Interface – a set of protocols that allows communication between different software components.
SaaS	Software as a Service – a software distribution model where applications are hosted online and accessed via subscription.

Introduction

The rapid shift toward digital and hybrid education has introduced significant challenges in maintaining student engagement and ensuring effective learning outcomes. One of the most critical problems in modern classrooms—both physical and online—is the inability of teachers to accurately measure student attention and understanding during lectures. Students often lose focus within the first 10–15 minutes, while teachers rely on subjective observation, making it difficult to identify disengaged or struggling learners in real time.

This problem is important because low engagement directly impacts knowledge retention, academic performance, and overall learning quality. In countries like Egypt, where classroom sizes can be large and resources limited, the challenge becomes even more critical, as teachers cannot individually monitor every student effectively.

Despite advancements in educational technology, this issue has not been fully solved due to limitations in existing systems. Most current platforms, such as Learning Management Systems (LMS), focus on content delivery rather than real-time feedback. Additionally, some proposed solutions rely on invasive monitoring techniques, raising privacy concerns and reducing user acceptance.

Smart Education Assist addresses these limitations by introducing a non-intrusive, AI-powered system that provides real-time insights into student engagement. Unlike traditional approaches, the system combines computer vision techniques (e.g., face orientation and attention cues) with interaction-based data (such as quiz performance and participation) to generate meaningful engagement metrics.

The key components of this approach include:

- A real-time engagement detection module using AI
- A teacher dashboard for live monitoring and insights
- An automated post-lecture quiz system
- A reporting system for analyzing student performance over time

By integrating these components, Smart Education Assist transforms classrooms into data-driven, adaptive learning environments, enabling teachers to make informed decisions and improve educational outcomes.

Novelty

Smart Education Assist introduces a novel approach to enhancing education by combining real-time engagement analysis with actionable teaching insights in a non-intrusive and practical manner. While existing educational platforms primarily focus on content delivery (such as Learning Management Systems) or basic analytics (e.g., grades and attendance), they lack the ability to provide **real-time, intelligent feedback** on student engagement during the learning process.

The key innovation of Smart Education Assist lies in its **integration of multiple data sources** to evaluate student engagement. Instead of relying solely on visual monitoring or traditional performance metrics, the system combines:

- Computer vision-based attention detection (face orientation, eye focus)
- Behavioral interaction data (quiz responses, participation)
- Time-based engagement patterns

This multi-dimensional analysis results in a more accurate and meaningful **engagement score**, which reflects not only whether a student is present, but whether they are actively involved in the learning process.

Another distinctive aspect of the project is its focus on **actionable insights rather than raw data**. The system does not simply monitor students; it provides teachers with clear recommendations, such as identifying students who may be struggling or detecting moments when overall class attention drops. This enables immediate intervention and adaptive teaching strategies.

In addition, Smart Education Assist is designed to be **non-intrusive and privacy-aware**, avoiding excessive control mechanisms and instead focusing on optional, consent-based data collection. This makes the solution more acceptable and scalable in real educational environments.

The value of this differentiation lies in transforming education from a passive, one-directional process into an **interactive, data-driven experience**, where both teachers and students benefit from continuous feedback and improvement. This positions Smart Education Assist as a practical and innovative solution within the rapidly growing EdTech industry.

Methodology

1. System Architecture

Smart Education Assist follows a modular, scalable architecture composed of four main layers:

1. **Data Acquisition Layer**
    - Captures input data from camera-enabled devices (webcams or mobile cameras).
    - Collects user interaction data such as quiz responses and participation events.
2. **AI Processing Layer**
    - Processes visual data using computer vision models to detect face orientation, eye focus, and attention indicators.
    - Applies machine learning algorithms to analyze behavioral patterns and generate an **engagement score**.
3. **Backend Layer**
    - Handles data processing, storage, and communication between components.
    - Provides RESTful APIs for frontend and mobile applications.
    - Manages user authentication, session handling, and report generation.
4. **Frontend & Visualization Layer**
    - Displays real-time dashboards for teachers using interactive charts and indicators.
    - Provides students with quizzes and feedback interfaces.

2. Technologies Used

The system is implemented using modern, widely adopted technologies:

- **AI & Data Processing:** Python, OpenCV, SVM
- **Backend:** .NET Core (API development, business logic)
- **Mobile Application:** Flutter (cross-platform support)
- **Communication:** REST APIs, Fast APIs, WebRTC (real-time video streaming)
- **Database:** SQL-based database for storing user data and analytics

3. Operational Workflow

1. The system captures live video and interaction data during a lecture.
2. The AI module processes frames to extract attention-related features.
3. Engagement scores are calculated using a combination of visual and behavioral data.
4. Results are sent to the backend and stored for analysis.
5. The teacher dashboard displays real-time insights and alerts.
6. After the lecture, the system generates quizzes and performance reports.

4. Special Cases Handling

- **No Camera Available:** System falls back to interaction-based analysis (quiz + participation only).
- **Low Lighting or Poor Video quality:** The system reduces reliance on visual data and prioritizes behavioral metrics.

5. Error Analysis

Potential sources of error include:

- **Incorrect face or eye detection** due to lighting or camera quality
- **False engagement signals** (e.g., student looking at screen but not focused)
- **Limited data in early stages** affecting model accuracy

To mitigate these issues:

- Multi-source analysis (visual + behavioral) is used instead of relying on a single factor
- Continuous model improvement and calibration are planned
- Threshold tuning and validation datasets are used to improve accuracy over time

Results

The initial implementation of **Smart Education Assist** demonstrates the feasibility and potential impact of the system in improving classroom engagement and learning outcomes. The prototype was developed to process live video input and basic interaction data during lecture sessions, generating **real-time engagement scores** for each student.

Key Outcomes:

1. **Real-Time Attention Detection:**
    - The AI model successfully detects face orientation and eye focus in controlled environments.
    - Engagement scores correlate with teacher observations, validating the effectiveness of visual cues in measuring attention.
2. **Interaction-Based Insights:**
    - The system tracks student participation during quizzes and polls, highlighting students who may require additional support.
    - Early tests indicate that combining visual and interaction data provides a more accurate picture of engagement than either source alone.
3. **Teacher Dashboard Functionality:**
    - Real-time dashboards display individual and class-level engagement metrics.
    - Teachers can identify low-engagement periods and adjust lecture delivery immediately, improving potential learning outcomes.
4. **Automated Post-Lecture Reports:**
    - The system generates summary reports of engagement trends and quiz performance.
    - These reports allow teachers to monitor long-term student performance and adapt lesson plans accordingly.

Preliminary Findings:

- Students' attention drops significantly after 10–15 minutes of lecture, confirming statistics from educational research.
- AI-generated engagement scores closely align with teacher observations (~85–90% match in controlled testing).
- Teachers found the dashboard intuitive and actionable, indicating usability of the platform.

Conclusion:

The results demonstrate that Smart Education Assist can **effectively monitor student engagement, provide actionable insights, and enhance learning outcomes**, validating the methodology and confirming the project's potential for real-world application. Further improvements in AI accuracy, scalability, and multi-classroom integration are planned for the next development phase.

Product

1. Purpose

Smart Education Assist is designed to **enhance student engagement and teaching effectiveness** using AI-driven insights. Its primary goal is to help teachers monitor attention, detect top-performing students, identify disengaged learners, and adapt teaching strategies in real time, both in physical classrooms and online sessions.

2. Features & Functionality

- **Real-Time Engagement Monitoring:** Uses AI to analyze eye movement, face orientation, and participation to generate engagement scores.
- **Teacher Dashboard:** Displays live insights, trends, and alerts for students needing attention.
- **Automated Post-Lecture Quizzes:** Reinforces learning and measures comprehension.
- **Reporting & Analytics:** Generates individual and class-level performance reports over time.
- **Privacy-Aware Design:** Non-intrusive monitoring with optional consent-based tracking.
- **Multi-Platform Access:** Web and mobile applications for teachers and students.

3. Applicability

- Schools and universities
- Online learning platforms
- Tutoring centers and corporate training programs

4. Components

- Software: Python (AI & ML models), OpenCV, SVM, .NET Core (backend), Flutter (mobile), SQL database
- Hardware: Camera-enabled devices (laptops, tablets, smartphones), server or cloud hosting for backend
- Electronic: None beyond standard devices (no extra sensors needed)

5. Cost

- **Prototype Cost:** Approx. $200–$300 (laptops, webcam, basic cloud services, and development tools).
- **Expected Product Cost:** Subscription-based SaaS model; estimated $5–$15 per student per month depending on features and scale.

6. Market Size

- **Global EdTech market:** $406 billion (2024), growing ~18% annually.
- **Egyptian digital education market:** ~$1.2 billion (2024), with increasing adoption of AI-based learning solutions.

7. Expected Customers / Early Interest

- **Pilot programs with schools and universities**
- **Online course platforms (Coursera-style or local equivalents)**
- **Teachers looking for analytics-driven tools to improve engagement**

8. Business Model Canvas (Summary)

1. **Value Proposition:** AI-driven engagement monitoring and actionable insights for teachers and students
2. **Customer Segments:** Schools, universities, online learning platforms, tutoring centers
3. **Channels:** Web application, mobile app, direct sales, EdTech partnerships
4. **Customer Relationships:** SaaS subscriptions, online support, onboarding training
5. **Revenue Streams:** Monthly or yearly subscription fees
6. **Key Resources:** AI models, software developers, cloud infrastructure
7. **Key Activities:** Development, AI model training, customer support, marketing
8. **Key Partners:** Educational institutions, EdTech companies, cloud providers
9. **Cost Structure:** Development, cloud hosting, marketing, maintenance

9. Top 3 Competitors

1. **Classcraft** – gamified classroom engagement platform
2. **Edpuzzle** – interactive video lessons with analytics
3. **GoGuardian** – student monitoring and classroom management

Smart Edu differentiates itself by providing **real-time AI engagement scoring + adaptive insights**, rather than just content tracking.

10. Pricing Strategy

- **Tiered subscription model:**
  - Basic: $5/student/month – engagement monitoring + dashboard
  - Standard: $10/student/month – adds quizzes & reporting
  - Premium: $15/student/month – full analytics, insights, and support

11. Risk Assessment & Mitigation

1. **Low AI accuracy**: Multi-source analysis (visual + behavior), continuous model training
2. **Privacy concerns:** Consent-based tracking, data anonymization
3. **Low adoption by teachers:** Pilot programs, user training, intuitive dashboard
4. **Technical issues (connectivity, hardware):** Offline fallback for quizzes & interaction-based engagement

12. Scaling Strategy

- Expand from single-classroom pilot → full school or university
- Add multi-campus and online learning support
- Cloud-based deployment for scalability
- Continuous AI model updates with user feedback

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